APEX LOW VOLTAGE · STRUCTURED CABLING · SECURITY

SOP-LV-031 — Electrical Safety — Working Near Energized Equipment

Minimum Precautions When Low-Voltage Work Meets Line-Voltage Infrastructure

SOP NUMBER	SOP-LV-031
TITLE	Electrical Safety — Working Near Energized Equipment
EFFECTIVE DATE	July 25, 2026
REVISION	1.0
APPLIES TO	All Apex field technicians — Ontario & Manitoba
CATEGORY	Electrical Safety

1. Purpose

This procedure establishes the minimum precautions Apex technicians follow whenever low-voltage work brings them into proximity with line-voltage electrical equipment — power supplies for maglocks, PoE injectors tied to building power, and panel areas near a low-voltage tie-in point.

2. Scope

This SOP applies to every Apex technician whose work requires access near an electrical panel, junction box, or line-voltage circuit. It draws a firm line around what Apex technicians do and do not do around energized equipment.

3. Definitions

- Energized — electrically connected to a source of voltage, or otherwise carrying voltage.
- De-energized / Isolated — disconnected from all sources of voltage and verified safe by a qualified person.
- Lockout/Tagout (LOTO) — the process of isolating and locking out an energy source so it cannot be re-energized while work is in progress.
- Qualified Electrical Worker — a licensed electrician authorized to work on energized or de-energized electrical systems.

4. Scope of Apex Work vs. a Licensed Electrician's Work

Apex technicians are low-voltage only. Apex technicians never open a live panel and never make line-voltage connections.

Any line-voltage work — tying a power supply into a dedicated circuit, opening a panel cover, resetting a breaker — is performed exclusively by a licensed electrician, either the client's or one contracted by Apex. Apex's role is coordinating safe access and confirming a circuit is de-energized before nearby low-voltage work proceeds.

5. Before Working Near an Open Panel

- Confirm with the licensed electrician on site that the relevant circuit has been locked out and tagged before you work nearby.
- If you are trained and equipped to do so, verify with a non-contact voltage tester — never rely on a breaker label or switch position alone.
- If there is no licensed electrician present and a panel needs to be opened for your work to proceed, stop and escalate to your supervisor.

6. Minimum Approach Distance

Stay clear of exposed energized parts at all times. There is no situation where an Apex technician needs to be within arm's reach of exposed live conductors — if a task seems to require it, stop and wait for the electrician to make the area safe first.

7. PPE for Proximity Work

- Safety glasses at minimum whenever working near an open panel, even de-energized.
- Non-conductive tools when working within reach of a panel area.
- No metal jewellery or watches when working near electrical equipment.

8. Prohibited Practices

- Never remove a panel cover.
- Never reset a tripped breaker.
- Never work on a circuit believed to be "hot," under any circumstance.
- Never assume a circuit is off because a switch is in the off position — always verify.

9. Incident Reporting

Any shock, arc flash, or electrical near-miss must be reported to your supervisor immediately and documented using the Apex Incident Report process. Any equipment involved is tagged out of service until inspected.

10. Regulatory References

- Ontario Electrical Safety Authority — Ontario Electrical Safety Code
- Ontario Occupational Health and Safety Act, s. 25 — general duty; O. Reg. 213/91, electrical hazard provisions
- CSA Z462 — Workplace Electrical Safety
- Manitoba Workplace Safety and Health Act, C.C.S.M. c. W210 — electrical safety provisions
- Manitoba Electrical Inspections — electrical safety requirements

11. Related Apex Documents

- SOP-LV-015 — Electric Strike & Maglock Installation
- SOP-LV-018 — Access Control Power Supply Sizing
- SOP-LV-029 — Lone Worker Safety

12. Revision History

Rev.	Date	Description	Approved By
1.0	Jul 25, 2026	Initial release	P. Cimini